
CAPSTONE PROJECT BLUEPRINT

Pancreatic Cancer Survival Prediction
Based on Preoperative Clinical Features

Fundamentals of Data Science — Capstone Project
University of the Philippines Manila
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Comprehensive Development Guide
From Project Framework to Final Submission

CONFIDENTIAL — INTERNAL PLANNING DOCUMENT

TABLE OF CONTENTS

PART I: PROJECT FRAMEWORK

1. Project Overview
2. Objectives
3. Scope and Constraints
4. Dataset Summary

PART II: TECHNICAL DECISIONS

5. Final Model Recommendations and Justifications
6. Preprocessing-to-Model Alignment
7. Evaluation Metrics

PART III: DEVELOPMENT PHASES

8. Phase 1 — Project Setup and Environment
9. Phase 2 — Data Loading and Initial Inspection
10. Phase 3 — Exploratory Data Analysis (EDA)
11. Phase 4 — Feature Engineering
12. Phase 5 — Feature Selection
13. Phase 6 — Train-Test Split
14. Phase 7 — Supervised Learning
15. Phase 8 — Unsupervised Learning
16. Phase 9 — Synthesis and Findings

PART IV: PRESENTATION AND SUBMISSION

17. Phase 10 — Presentation Creation
18. Phase 11 — Final Review and Submission
19. Presentation Slide Content Outline (17 Slides)
20. Interpretation and Reasoning Guide (Professor Q&A; Prep)
21. Common Pitfalls to Avoid
22. Visualizations Master List (16 Required)
23. Timeline and Task Allocation

PART I: PROJECT FRAMEWORK

1. Project Overview

This capstone project applies foundational data science techniques to a real-world clinical dataset: the PCSPF (Pancreatic Cancer Survival based on Preoperative Features) dataset containing 878 patient records with 20 usable preoperative clinical features. The central task is a binary classification problem predicting whether a pancreatic cancer patient will survive one year or more (label = 1) or less than one year (label = 0) after diagnosis, using only measurements available before surgery.

The project requires the implementation of exactly one supervised learning technique and one unsupervised learning technique on this dataset. Beyond simply building models, the project demands a complete data science pipeline: from exploratory analysis and preprocessing, through model development and evaluation, to interpretation, synthesis, and professional presentation of findings. The clinical context is essential to the project's value. Pancreatic cancer has one of the lowest survival rates among all cancers, and the ability to stratify patients by predicted survival using routine preoperative blood tests and markers could inform treatment planning and resource allocation.

2. Objectives

Primary Objectives

- Build a supervised classification model capable of predicting 1-year pancreatic cancer survival with performance meaningfully above the majority-class baseline of 68.8% accuracy
- Apply an unsupervised clustering technique to discover natural patient subgroups within the preoperative feature space, independent of the survival label
- Synthesize supervised and unsupervised findings to produce a unified analytical narrative connecting predictive modeling with pattern discovery

Secondary Objectives

- Demonstrate mastery of the end-to-end data science pipeline (data exploration, preprocessing, modeling, evaluation, interpretation)
- Produce clinically interpretable results that explain which preoperative features are most informative for survival prediction
- Deliver a professional-quality written report, reproducible codebase, and oral presentation/defense
- Identify and document the limitations of the analysis and suggest concrete directions for future work

3. Scope and Constraints

In Scope

- One supervised technique: Random Forest Classifier (recommended)
- One unsupervised technique: K-Means Clustering (recommended)

- Full EDA, preprocessing, feature engineering, feature selection, model evaluation, and interpretation
- Professional documentation (report, code, slides) and oral defense preparation

Out of Scope

- Deep learning or neural network approaches (not appropriate for a Fundamentals course with only 878 samples)
- External dataset augmentation or web-scraped supplementary data
- Clinical validation or deployment of the model in a medical setting
- Multiple supervised or unsupervised models (the requirement is exactly one of each)

Known Constraints

- **Moderate class imbalance:** 604 positive (68.8%) vs. 274 negative (31.2%); must be explicitly addressed
- **Weak individual feature correlations:** No single feature has a correlation with the target exceeding [0.15]; the model must capture feature interactions
- **Multicollinearity among derived ratios:** NLR, PLR, SII, and CRP/ALB are mathematically derived from other features in the dataset
- **Anonymized IDs and empty prediction columns:** Three columns (ID, Predict label 1, Predict label 0) are unusable and must be dropped

4. Dataset Summary

The PCSPF dataset contains 878 rows (patients) and 24 columns. After removing 3 unusable columns, 20 features remain for analysis alongside the binary target variable.

Feature Inventory

Feature Name	Type	Description
Sex	Binary (0/1)	Patient sex (0 = Female, 1 = Male)
Abdominal Pain	Binary (0/1)	Presence of abdominal pain symptom
Age	Continuous	Patient age at diagnosis
BMI	Continuous	Body Mass Index
CRP	Continuous	C-Reactive Protein (inflammation marker)
ALB	Continuous	Albumin (nutritional/liver status)
CRP/ALB	Continuous	CRP-to-Albumin ratio (derived)
Leukocyte	Continuous	White blood cell count
Neutrocyte	Continuous	Neutrophil count
Platelet	Continuous	Platelet count
Lymphocyte	Continuous	Lymphocyte count

Feature Name	Type	Description
NLR	Continuous	Neutrophil-to-Lymphocyte Ratio (derived)
PLR	Continuous	Platelet-to-Lymphocyte Ratio (derived)
SII	Continuous	Systemic Immune-Inflammation Index (derived)
Lactic Dehydrogenase	Continuous	LDH enzyme level
CA19-9	Continuous	Carbohydrate Antigen 19-9 (tumor marker)
CEA	Continuous	Carcinoembryonic Antigen (tumor marker)
Prealbumin	Continuous	Prealbumin (nutritional status)
Total Bilirubin	Continuous	Total bilirubin level
Directed Bilirubin	Continuous	Direct bilirubin level

Dropped Columns

Column	Reason for Removal
ID	Anonymized as 'xxx' for all 878 rows; carries zero information
Predict label 1	Entirely NaN (all 878 values missing); placeholder from original study
Predict label 0	Entirely NaN (all 878 values missing); placeholder from original study

Target Variable Distribution

Class	Label Meaning	Count	Percentage
1	Survived $\geq$ 1 year	604	68.8%
0	Survived $<$ 1 year	274	31.2%

Note: The 2.2:1 class ratio means a naive majority-class classifier achieves 68.8% accuracy. Your model must meaningfully exceed this baseline, and evaluation must include metrics beyond accuracy.

PART II: TECHNICAL DECISIONS

5. Final Model Recommendations and Justifications

Supervised: Random Forest Classifier

Random Forest is recommended as the supervised technique for the following reasons:

- **Handles mixed feature types naturally:** The dataset contains both binary (Sex, Abdominal Pain) and continuous features; tree-based methods require no special encoding or scaling for this mix
- **Captures non-linear feature interactions:** Since individual correlations with survival are weak (all below $|0.15|$), a model that can discover complex interactions between features is essential. Random Forest excels at this through its ensemble of decision trees
- **Robust to multicollinearity:** The derived ratio features (NLR, PLR, SII, CRP/ALB) introduce mathematical dependencies among features. Random Forest's random feature subsampling at each split mitigates the impact of correlated predictors
- **Built-in feature importance:** Provides interpretable rankings of which preoperative features contribute most to survival prediction, which is directly useful for clinical interpretation
- **Resistant to overfitting:** The ensemble averaging across hundreds of trees provides natural regularization, important given the moderate dataset size of 878 samples
- **No feature scaling required:** Tree-based methods are invariant to monotonic transformations of features, simplifying the preprocessing pipeline for the supervised component

Unsupervised: K-Means Clustering

K-Means Clustering is recommended as the unsupervised technique for the following reasons:

- **Discovers natural patient subgroups:** K-Means partitions patients into groups based on similarity in their clinical profiles, potentially revealing clinically meaningful subtypes (e.g., high-inflammation vs. low-inflammation patients)
- **Straightforward and interpretable:** Cluster centroids directly represent the average clinical profile of each subgroup, making results easy to explain to both technical and non-technical audiences
- **Strong visual output:** When reduced to 2D via PCA, cluster assignments produce clear, intuitive scatter plots that are effective in presentations
- **Bridges naturally with the supervised analysis:** After clustering (without using labels), overlaying survival rates per cluster provides a powerful synthesis point connecting unsupervised structure to the prediction task
- **Well-supported model selection methods:** The Elbow Method and Silhouette Score provide principled, visual approaches to choosing the number of clusters, demonstrating methodological rigor

Why Not Other Techniques?

Technique	Category	Reason Not Chosen
Logistic Regression	Supervised	Assumes linear relationships; weak individual correlations suggest non-linear patterns matter more

Technique	Category	Reason Not Chosen
SVM	Supervised	Less interpretable (no direct feature importance); harder to tune and explain in a defense
XGBoost / Gradient Boosting	Supervised	More complex to justify in a Fundamentals course; marginal gains unlikely to offset added complexity
DBSCAN	Unsupervised	Requires careful epsilon/min_samples tuning; may find few or no clusters in high-dimensional continuous data
Hierarchical Clustering	Unsupervised	Computationally expensive at n=878; dendrogram interpretation is less intuitive for clinical bridging
PCA (alone)	Unsupervised	Dimensionality reduction, not clustering; does not produce patient subgroups directly

6. Preprocessing-to-Model Alignment

Different techniques have different preprocessing requirements. The table below maps each step to where it applies, ensuring nothing is misapplied.

Preprocessing Step	Random Forest	K-Means	Rationale
Drop ID, Predict label 1/0	Yes	Yes	Unusable columns for both techniques
StandardScaler (z-score normalization)	Not needed	Required	K-Means uses Euclidean distance, which is scale-sensitive; RF is scale-invariant
Train-test split (80/20, stratified)	Yes	No (use full data)	Supervised needs held-out evaluation; unsupervised explores full population structure
Class imbalance handling (SMOTE or class_weight)	Yes	Not applicable	Only relevant where the target label is used during training
Feature selection / dropping derived ratios	Optional (RF handles multicollinearity)	Recommended (reduces redundancy in distance calculations)	Multicollinearity affects distance-based methods more than tree-based ones

Note: Critical data leakage rule: StandardScaler must be fit on the training set only, then transform applied to both training and test sets. Never fit the scaler on the full dataset before splitting.

7. Evaluation Metrics

Supervised Model Evaluation

Metric	What It Measures	Why It Matters Here
Accuracy	Overall correct predictions / total	Baseline is 68.8% (majority class); must exceed this meaningfully
Precision (Class 0)	Of predicted deaths < 1yr, how many were correct	False positives (predicting death when patient survives) cause unnecessary alarm
Recall (Class 0)	Of actual deaths < 1yr, how many were caught	Missing high-risk patients (false negatives) is clinically dangerous
F1-Score	Harmonic mean of precision and recall	Balances both error types; especially important for the minority class
AUC-ROC	Area under Receiver Operating Characteristic curve	Measures discrimination ability across all thresholds; threshold-independent
Confusion Matrix	Full 2x2 breakdown of TP, TN, FP, FN	Provides complete picture; essential for clinical interpretation

Unsupervised Model Evaluation

Metric	What It Measures	Why It Matters Here
Inertia (Elbow Method)	Within-cluster sum of squared distances to centroid	Identifies the point of diminishing returns for adding more clusters
Silhouette Score	How similar a point is to its cluster vs. nearest other cluster	Values range from -1 to 1; higher means better-defined clusters
Cluster-Survival Cross-tabulation	Survival rate per cluster (using labels post-hoc)	The key bridge metric: do discovered clusters align with different outcomes?

PART III: DEVELOPMENT PHASES

8. Phase 1 — Project Setup and Environment

Set up the development environment and establish project structure before writing any analysis code.

Tasks

- Install Python 3.9+ and Jupyter Notebook (or JupyterLab)
- Create a virtual environment and install required libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, imbalanced-learn (for SMOTE), openpyxl (for .xlsx reading)
- Create project folder structure:
 - **data/** — raw dataset file
 - **notebooks/** — Jupyter notebooks for development
 - **figures/** — saved plot images for the report
 - **report/** — written report drafts and final PDF
 - **slides/** — presentation file
- Verify all libraries import correctly by running a test cell
- Place the PCSPF Excel file in the data/ directory

Deliverable

- A working, organized project environment ready for development

9. Phase 2 — Data Loading and Initial Inspection

Load the dataset and perform a first-pass sanity check to confirm the data matches expectations.

Tasks

- Load the Excel file using pandas (`pd.read_excel`)
- Print and verify: shape (expected: 878 rows, 24 columns)
- Print column names, data types (`dtypes`), and first/last 5 rows (`head/tail`)
- Check for missing values using `.isnull().sum()` — confirm: 0 nulls in all usable features, 878 nulls in Predict label 1 and Predict label 0
- Check for duplicate rows using `.duplicated().sum()`
- Verify target variable distribution using `.value_counts()` — confirm: 604 vs. 274
- Identify and document the 3 columns to drop (ID, Predict label 1, Predict label 0) and drop them, resulting in a 878 x 21 DataFrame (20 features + 1 target)

Deliverable

- A cleaned DataFrame (878 x 21) with all unusable columns removed, ready for EDA
- A summary table of the initial inspection findings (to include in the report's Dataset Description section)

10. Phase 3 — Exploratory Data Analysis (EDA)

Explore the dataset visually and statistically to understand distributions, relationships, and potential issues before modeling.

Tasks

- **Summary statistics:** Generate `.describe()` for all features and format as a table for the report
- **Target distribution plot:** Bar chart showing the count of class 0 vs. class 1 with percentage annotations
- **Feature distributions:** Histograms for all 18 continuous features (use a 4x5 or 3x6 grid subplot). Note any heavily skewed features (likely CA19-9, CRP, bilirubin)
- **Boxplots by survival:** Side-by-side boxplots for each continuous feature, grouped by the target variable, to visually identify features that differ between classes
- **Correlation heatmap:** Full correlation matrix of all 20 features plus the target, using seaborn heatmap with annotation. Identify strong inter-feature correlations (especially among derived ratios)
- **Multicollinearity analysis:** Explicitly note and discuss the mathematical dependencies: $NLR = \text{Neutrocyte} / \text{Lymphocyte}$, $PLR = \text{Platelet} / \text{Lymphocyte}$, $SII = \text{Platelet} \times \text{Neutrocyte} / \text{Lymphocyte}$, $CRP/ALB = \text{CRP} / \text{ALB}$
- **Class-conditional means:** Compute mean of each feature for class 0 vs. class 1 and present as a comparison table. Highlight features with the largest differences
- **Outlier identification:** Note features with extreme outliers (via boxplots or IQR method). Document whether you choose to keep or remove them and justify your decision

Deliverable

- At least 5 publication-quality visualizations (target distribution, histograms, boxplots, correlation heatmap, class-conditional comparison)
- Written EDA summary (2–3 paragraphs) identifying key findings, to be included in the report

11. Phase 4 — Feature Engineering

Create or transform features that may improve model performance or interpretability.

Tasks

- **Assess whether new features are needed:** Since the dataset already includes derived ratios (NLR, PLR, SII, CRP/ALB), additional engineering may not be necessary. Document this decision explicitly
- **If creating new features** (optional, not required):
 - Age binning (e.g., <50, 50–65, 65+) for interpretability
 - BMI categorization (underweight, normal, overweight, obese) based on WHO thresholds
 - Log-transformation of heavily right-skewed features (CA19-9, CRP, bilirubin) to reduce outlier influence for K-Means
- **Document all transformations** in the Methodology section of the report, including the rationale for each

Deliverable

- Final engineered feature set with justification for all decisions (including the decision not to engineer if applicable)

Note: Feature engineering is optional for this project. The existing features are sufficient. If you do engineer features, keep it minimal and well-justified. Over-engineering without clinical rationale weakens your defense.

12. Phase 5 — Feature Selection

Determine the final set of features to feed into each model, addressing multicollinearity concerns.

Tasks

- **For Random Forest:** You may keep all 20 features. Random Forest handles multicollinearity through its random feature subsampling mechanism. Document that you are aware of the collinearity and explain why it is acceptable for this model
- **For K-Means:** Consider dropping derived ratio features to reduce redundancy in the distance-based clustering. Two approaches:
 - **Option A (Recommended):** Keep all features for K-Means as well, and note the multicollinearity as a limitation. This is simpler and defensible
 - **Option B:** Drop NLR, PLR, SII, and CRP/ALB (keep their component features instead). Justify by explaining that Euclidean distance double-counts the information embedded in these ratios
- **Whichever option you choose, document and justify it in the report**
- **Compute Variance Inflation Factor (VIF)** for all features as a quantitative measure of multicollinearity (optional but impressive). VIF > 10 indicates high multicollinearity

Deliverable

- A clearly documented feature selection decision for both the supervised and unsupervised models
- (Optional) VIF table showing multicollinearity levels

13. Phase 6 — Train-Test Split

Split the data for supervised model training and evaluation. This step applies only to the supervised pipeline.

Tasks

- Separate features (X) and target (y) from the cleaned DataFrame
- Apply an 80/20 train-test split using sklearn's `train_test_split` with `stratify=y` to preserve class proportions in both sets
- Set `random_state=42` (or any fixed seed) for reproducibility
- Verify class proportions in train and test sets match the original (~68.8% / 31.2%)
- If using `StandardScaler` for any purpose, fit it on the training set only, then transform both train and test
- If using SMOTE for class imbalance, apply it only to the training set after splitting (never SMOTE the test set)

Critical Data Leakage Rules

- **Never scale before splitting.** Fitting a scaler on the full dataset leaks test-set statistics into training
- **Never SMOTE before splitting.** Synthetic samples from the full dataset may duplicate test-set information into training
- **The test set must remain untouched** until final evaluation

Deliverable

- X_train, X_test, y_train, y_test arrays ready for modeling
- Confirmed class proportions in both splits (include a brief verification print in the notebook)

14. Phase 7 — Supervised Learning (Random Forest Classifier)

Train, tune, evaluate, and interpret the Random Forest model.

Step 7.1: Baseline Model

- Instantiate RandomForestClassifier with class_weight='balanced' to address class imbalance (alternative: apply SMOTE to training data)
- Use default hyperparameters initially: n_estimators=100, max_depth=None, min_samples_split=2
- Fit on training data (X_train, y_train)
- Predict on X_test and compute initial accuracy, F1-score, and confusion matrix

Step 7.2: Hyperparameter Tuning

- Define a parameter grid for tuning:
 - n_estimators: [100, 200, 300]
 - max_depth: [None, 10, 15, 20]
 - min_samples_split: [2, 5, 10]
 - min_samples_leaf: [1, 2, 4]
- Run GridSearchCV or RandomizedSearchCV with 5-fold stratified cross-validation, scoring='f1' (prioritize F1 over accuracy due to imbalance)
- Record best parameters and best cross-validation score
- Refit the best model on full training data

Step 7.3: Final Evaluation

- Predict on the held-out test set using the tuned model
- Generate and record all evaluation metrics:
 - Accuracy (and comparison to 68.8% baseline)
 - Classification Report (precision, recall, F1 for both classes)
 - Confusion Matrix (visualized as a heatmap with seaborn)
 - ROC Curve with AUC score (plotted and annotated)
- Save all metric values in a summary table for the report

Step 7.4: Feature Importance

- Extract feature_importances_ from the trained model
- Plot a horizontal bar chart of all features ranked by importance (most important at top)
- Identify and discuss the top 5 most important features
- Relate feature importances to clinical knowledge (e.g., 'CA19-9 is a well-known pancreatic tumor marker, and its high importance aligns with medical literature')

Deliverable

- Trained and tuned Random Forest model with all evaluation metrics computed and recorded
- Confusion matrix heatmap, ROC curve, and feature importance bar chart saved as figures
- Written interpretation of results (1–2 paragraphs) for the report

15. Phase 8 — Unsupervised Learning (K-Means Clustering)

Discover natural patient subgroups using K-Means, then connect findings to survival outcomes.

Step 8.1: Data Preparation for Clustering

- Use the full dataset (all 878 patients) without the target column for clustering
- Apply StandardScaler to all features (K-Means requires scaled features because it uses Euclidean distance)
- If you chose to drop derived ratio features in Phase 5, apply that here

Step 8.2: Optimal Cluster Selection

- Run K-Means for $k = 2$ through $k = 10$ and record inertia for each
- Plot the Elbow Method graph (inertia vs. k) and identify the 'elbow' point
- Compute Silhouette Scores for $k = 2$ through $k = 10$ and plot
- Select the optimal k based on both methods (likely $k = 2, 3$, or 4)
- Document your choice and reasoning

Step 8.3: Fit Final Clustering Model

- Fit K-Means with the chosen k on the scaled data
- Assign cluster labels to each patient
- Apply PCA ($n_components=2$) to reduce the scaled feature space to 2 dimensions for visualization
- Create a 2D scatter plot colored by cluster assignment, with cluster centroids marked

Step 8.4: Cluster Profiling

- Compute the mean of each original (unscaled) feature per cluster and present as a table or heatmap
- Identify and describe what distinguishes each cluster clinically (e.g., 'Cluster 1: older patients with high CRP and low albumin, suggesting systemic inflammation')
- Count the number of patients in each cluster

Step 8.5: Cluster-Survival Bridge Analysis

- Cross-tabulate cluster assignments against the actual survival label
- Compute the survival rate (percentage of label = 1) within each cluster
- Visualize as a grouped or stacked bar chart (cluster on x-axis, survival proportion on y-axis)
- Perform a chi-square test of independence between cluster assignment and survival label to test statistical significance
- Interpret: do different clusters have meaningfully different survival rates?

Deliverable

- Elbow plot, silhouette plot, PCA cluster scatter, cluster profile heatmap, and cluster-survival bar chart saved as figures
- Cluster profiling table with mean feature values per cluster
- Written interpretation connecting clusters to clinical profiles and survival (1–2 paragraphs)

16. Phase 9 — Synthesis and Findings

Connect the supervised and unsupervised results into a unified narrative. This is the analytical core that demonstrates data science thinking beyond just running models.

Tasks

- **Feature importance vs. cluster differentiation:** Compare the top features from Random Forest's importance ranking with the features that most differentiate the clusters. Do they overlap? If CA19-9 is the most important predictor and also the feature with the largest difference between clusters, that is a strong converging finding
- **High-risk cluster identification:** Identify which cluster(s) have the lowest survival rates and describe their clinical profile in plain language
- **Model predictions vs. clusters:** Optionally, apply the Random Forest model to the full dataset to get predicted probabilities, then compute the average predicted survival probability per cluster. Do the model's predictions align with the cluster-level survival rates?
- **Clinical narrative:** Write a synthesis paragraph such as: 'Cluster 2, characterized by elevated CA19-9, high neutrophil-to-lymphocyte ratio, and low prealbumin, exhibited the lowest 1-year survival rate (X%). These same features ranked among the top 5 predictors in the Random Forest model, suggesting that systemic inflammation and tumor burden are the primary drivers of poor prognosis in this cohort.'
- **Limitations:** Document at least 3–5 limitations:
 - Moderate class imbalance may bias predictions toward the majority class despite mitigation efforts
 - Weak individual feature correlations limit single-feature interpretability
 - No external validation set; results may not generalize to other populations
 - K-Means assumes spherical clusters with equal variance, which may not reflect true data structure
 - Derived ratio features introduce redundancy that may inflate apparent feature importance
- **Future work suggestions:**
 - Test additional classifiers (e.g., Gradient Boosting, SVM) for comparison
 - Apply DBSCAN or hierarchical clustering as alternative unsupervised methods
 - Collect post-operative features for a more comprehensive prediction model
 - Validate on an independent external cohort

Deliverable

- A written synthesis section (2–3 paragraphs) connecting supervised and unsupervised findings
- Completed Limitations and Future Work sections for the report

PART IV: PRESENTATION AND SUBMISSION

17. Phase 10 — Presentation Creation

Prepare a professional slide deck for the oral defense. The presentation should tell a concise story, not replicate the full report.

Guidelines

- Target 17 slides (detailed outline in Section 19 below)
- Keep text minimal on slides; use visuals (plots, tables, diagrams) as the primary content
- Every slide should have a clear takeaway; avoid slides that are purely decorative or transitional without content
- Prepare speaker notes for each slide with talking points and anticipated questions
- Practice the full presentation to fit within 15–20 minutes (adjust to professor's time allocation)
- Prepare a 1–2 minute introduction that frames the clinical problem before diving into technical details

18. Phase 11 — Final Review and Submission

Perform a comprehensive quality check before submission.

Report Checklist

- All sections present: Title, Abstract, Introduction, Dataset Description, Methodology, Results, Discussion, Conclusion, References, Appendix
- All figures have axis labels, titles, legends, and figure numbers/captions
- All tables have headers and are referenced in the text
- Consistent terminology throughout (e.g., always 'Random Forest' not sometimes 'RF' and sometimes 'random forest')
- Proofread for grammar, spelling, and flow
- References cited for scikit-learn, imbalanced-learn, dataset source, and any clinical claims

Code Checklist

- Notebook runs top-to-bottom without errors (Kernel > Restart & Run All)
- All cells are in logical order with section headers (Markdown cells)
- No leftover debug prints, commented-out experiments, or unused imports
- All `random_state` values are fixed for reproducibility
- All figures saved to files using `plt.savefig()` with `dpi=300` and `bbox_inches='tight'`

Submission Package

- `capstone_report.pdf` (or `.docx`)
- `capstone_notebook.ipynb` (clean, reproducible)

- capstone_slides.pptx (or .pdf)
- data/ folder with the original dataset
- figures/ folder with all saved plots
- README.txt (optional but professional: brief description of files and how to run the notebook)

19. Presentation Slide Content Outline (17 Slides)

- **Slide 1: Title Slide** — Project title, team members, course name, professor, date, university logo/branding
- **Slide 2: Agenda / Outline** — Brief roadmap of the presentation (6–7 bullet items corresponding to the major sections)
- **Slide 3: Problem Statement** — What is pancreatic cancer? Why is survival prediction important? Frame the clinical need in 2–3 sentences, supported by a key statistic (e.g., 5-year survival rate ~12%)
- **Slide 4: Objectives** — List the 3–4 primary objectives clearly. One for supervised, one for unsupervised, one for synthesis, one for deliverables
- **Slide 5: Dataset Overview** — Source, size (878 patients, 20 features), feature categories (demographics, inflammation, tumor markers, nutritional), target distribution bar chart
- **Slide 6: EDA Highlights** — 2–3 key EDA visualizations: correlation heatmap (showing multicollinearity), boxplots of top discriminating features (e.g., CA19-9, Abdominal Pain), class-conditional means
- **Slide 7: Methodology Overview** — Pipeline diagram or flowchart: Data → EDA → Preprocessing → Unsupervised → Supervised → Synthesis. Mention techniques by name
- **Slide 8: Preprocessing Summary** — Dropped columns, scaling decisions, train-test split details, class imbalance strategy. Keep to a concise table format
- **Slide 9: K-Means: Cluster Selection** — Elbow Method plot and Silhouette Score plot side by side. State the chosen k and why
- **Slide 10: K-Means: Cluster Visualization** — PCA 2D scatter plot colored by cluster. Brief description of what each cluster represents
- **Slide 11: K-Means: Cluster Profiles** — Cluster profiling heatmap or table showing mean feature values per cluster. Highlight distinguishing features
- **Slide 12: K-Means: Survival Overlay** — Cluster-vs-survival bar chart. State survival rate per cluster. Note chi-square test result if significant
- **Slide 13: Random Forest: Model Performance** — Confusion matrix heatmap and classification report table. State accuracy vs. baseline (68.8%), F1-score, and AUC
- **Slide 14: Random Forest: ROC Curve** — ROC curve with AUC annotated. Briefly explain what AUC represents for any non-technical audience members
- **Slide 15: Random Forest: Feature Importance** — Feature importance bar chart (top 10 or all features). Highlight top 3–5 and relate to clinical knowledge
- **Slide 16: Synthesis and Key Findings** — The core slide: how supervised and unsupervised results connect. Use a side-by-side or table format comparing top RF features with cluster-distinguishing features. State the clinical narrative in 2–3 sentences
- **Slide 17: Conclusion, Limitations, and Future Work** — 3 key takeaways, 3–5 limitations, 2–3 future work directions. End with a Thank You and Q&A; prompt

20. Interpretation and Reasoning Guide (Professor Q&A; Prep)

Prepare answers for the most likely defense questions. Below are 15 common questions with guidance on how to answer them.

Q: "Why did you choose Random Forest over other classifiers?"

Guidance: Because it handles mixed feature types, captures non-linear interactions (important given weak individual correlations), is robust to multicollinearity from derived ratios, provides built-in feature importance for clinical interpretation, and resists overfitting through ensemble averaging.

Q: "Why K-Means and not DBSCAN or hierarchical clustering?"

Guidance: K-Means is interpretable (centroids represent average clinical profiles), has principled model selection methods (Elbow, Silhouette), produces clear visual output via PCA, and bridges naturally with the supervised analysis by overlaying survival rates per cluster.

Q: "How did you handle class imbalance?"

Guidance: We used `class_weight='balanced'` in Random Forest, which adjusts the loss function to penalize misclassification of the minority class more heavily. Alternative: SMOTE was applied to the training set only. We evaluated using F1 and AUC-ROC rather than relying on accuracy alone.

Q: "Why not use accuracy as your primary metric?"

Guidance: Because the majority class is 68.8%. A model that predicts all patients as 'survive ≥ 1 year' achieves 68.8% accuracy while being clinically useless. F1-score and AUC-ROC better capture the model's ability to identify the minority (high-risk) class.

Q: "What does your confusion matrix tell you?"

Guidance: It shows the trade-off between catching high-risk patients (recall for class 0) and avoiding false alarms (precision for class 0). Discuss the specific numbers: how many class-0 patients were correctly identified vs. missed.

Q: "What is the clinical significance of your top features?"

Guidance: CA19-9 is a well-established pancreatic tumor marker; elevated levels indicate aggressive disease. NLR and SII reflect systemic inflammation, a known prognostic factor. Abdominal pain presence indicates more advanced disease. Prealbumin reflects nutritional status, which affects treatment tolerance.

Q: "Why did you scale features for K-Means but not Random Forest?"

Guidance: K-Means uses Euclidean distance to assign points to clusters. Without scaling, features with larger numeric ranges (e.g., CA19-9 can reach thousands) would dominate the distance calculation. Random Forest splits on individual features and is invariant to monotonic transformations.

Q: "How did you choose the number of clusters?"

Guidance: We used the Elbow Method (looking for the point where adding more clusters yields diminishing reductions in inertia) and the Silhouette Score (choosing k with the highest average silhouette). Both methods converged on $k =$ [your result].

Q: "What do your clusters represent clinically?"

Guidance: Describe each cluster by its distinguishing features in plain language. E.g., 'Cluster 1 represents younger patients with lower inflammatory markers and normal tumor marker levels, while Cluster 3 represents

older patients with elevated CA19-9 and systemic inflammation.'

Q: "How do your supervised and unsupervised results connect?"

Guidance: The features that Random Forest identified as most important for predicting survival (e.g., CA19-9, NLR, Abdominal Pain) are the same features that most differentiate the clusters. The cluster with the worst clinical profile (highest inflammation, highest tumor markers) also had the lowest survival rate, independently confirming the supervised model's findings.

Q: "What are the limitations of your analysis?"

Guidance: Moderate class imbalance despite mitigation, weak individual correlations, no external validation, K-Means assumes spherical clusters, derived ratios introduce redundancy, and the dataset is from a single institution, limiting generalizability.

Q: "What would you do differently with more time?"

Guidance: Test additional classifiers for comparison (XGBoost, SVM), apply alternative clustering (DBSCAN, hierarchical), perform more rigorous feature selection (recursive elimination), validate on an external dataset, and explore survival analysis methods (Cox regression, Kaplan-Meier) as complements.

Q: "Why did you keep/drop the derived ratio features?"

Guidance: If kept: Random Forest handles multicollinearity through random feature subsampling, and these ratios are clinically meaningful in their own right. If dropped for K-Means: Euclidean distance double-counts the information from component features, distorting cluster assignments.

Q: "How do you know your model is not overfitting?"

Guidance: We used a held-out test set that the model never saw during training or tuning. We also used cross-validation during hyperparameter tuning. The test set performance is comparable to the cross-validation performance, indicating the model generalizes.

Q: "Could your results be used clinically?"

Guidance: Not in their current form. This is a retrospective analysis on a single dataset without external validation. However, the findings support the hypothesis that routine preoperative blood tests can contribute to risk stratification, which could inform future prospective studies.

21. Common Pitfalls to Avoid

Pitfall	Why It Matters
Data Leakage	Fitting the scaler on the full dataset before train-test split, or applying SMOTE before splitting. Both leak information from the test set into training.
Using accuracy as the sole metric	With 68.8% majority class, accuracy is misleading. Always pair with F1, AUC-ROC, and confusion matrix.
Ignoring class imbalance entirely	Training a model without <code>class_weight='balanced'</code> or SMOTE will bias it toward predicting the majority class.
Scaling features for Random Forest	Not harmful, but unnecessary and shows a misunderstanding of tree-based models. If asked why, you should know the answer.
Not scaling features for K-Means	This is a critical error. Unscaled K-Means will be dominated by high-range features like CA19-9.
Using the target label in unsupervised analysis	K-Means must be performed without the survival label. The label is only used post-hoc to evaluate whether clusters align with outcomes.
Treating the Predict label columns as usable features	Both are entirely NaN. Attempting to use them will cause errors or introduce meaningless noise.
Over-engineering features without justification	Adding many derived features without clinical or statistical rationale weakens your defense.
Not explaining why you chose your techniques	Simply saying 'I used Random Forest' without a justification is insufficient for a capstone defense.
Copy-pasting code without understanding	If asked to explain any line of code in your notebook, you must be able to do so.
Forgetting to set <code>random_state</code>	Without fixed seeds, results are non-reproducible, which is unacceptable in a research context.
Making clinical claims without hedging	Say 'the model suggests' or 'the data indicates,' not 'this proves' or 'this confirms.'

22. Visualizations Master List (16 Required)

Every visualization listed below should be produced, saved as a high-resolution image (dpi=300), and included in both the report and presentation where indicated.

#	Visualization	Chart Type	Phase	Where Used
1	Target Class Distribution	Bar chart	EDA	Report + Slide 5
2	Feature Histograms (all 18 continuous)	Grid of histograms	EDA	Report

#	Visualization	Chart Type	Phase	Where Used
3	Boxplots by Survival Label	Grouped boxplots	EDA	Report + Slide 6
4	Correlation Heatmap	Annotated heatmap	EDA	Report + Slide 6
5	Class-Conditional Feature Means	Grouped bar chart or table	EDA	Report
6	Elbow Method Plot	Line plot (inertia vs. k)	Unsupervised	Report + Slide 9
7	Silhouette Score Plot	Line plot (silhouette vs. k)	Unsupervised	Report + Slide 9
8	PCA Cluster Scatter Plot	2D scatter (colored by cluster)	Unsupervised	Report + Slide 10
9	Cluster Profile Heatmap	Heatmap (mean features per cluster)	Unsupervised	Report + Slide 11
10	Cluster-Survival Cross-tabulation	Grouped/stacked bar chart	Unsupervised	Report + Slide 12
11	Confusion Matrix	Heatmap (seaborn)	Supervised	Report + Slide 13
12	ROC Curve with AUC	Line plot with diagonal reference	Supervised	Report + Slide 14
13	Feature Importance Bar Chart	Horizontal bar chart	Supervised	Report + Slide 15
14	Classification Report Table	Formatted table (precision, recall, F1)	Supervised	Report + Slide 13
15	PCA Scatter Colored by Survival Label	2D scatter (colored by actual label)	Synthesis	Report + Slide 16
16	Synthesis Comparison (RF importance vs. cluster features)	Side-by-side chart or table	Synthesis	Report + Slide 16

23. Timeline and Task Allocation

Below is a suggested 10-day work plan assuming a solo developer. Adjust based on team size and deadline.

Day	Tasks	Milestone
Day 1	Project Setup (Phase 1) + Data Loading (Phase 2)	Environment ready, data cleaned, initial inspection complete
Day 2	Exploratory Data Analysis (Phase 3)	All EDA visualizations produced, summary written
Day 3	Feature Engineering + Selection (Phases 4–5)	Final feature sets defined for both models

Day	Tasks	Milestone
Day 4	Train-Test Split + Unsupervised Learning (Phases 6, 8 Steps 8.1–8.3)	Data split, K-Means fitted, clusters visualized
Day 5	Unsupervised Learning continued (Phase 8 Steps 8.4–8.5)	Cluster profiling complete, survival overlay done
Day 6	Supervised Learning (Phase 7 Steps 7.1–7.2)	Baseline model trained, hyperparameter tuning complete
Day 7	Supervised Learning continued (Phase 7 Steps 7.3–7.4) + Synthesis (Phase 9)	All evaluation metrics computed, synthesis narrative written
Day 8	Report Writing	Full draft of report complete
Day 9	Presentation Creation (Phase 10)	All 17 slides created with speaker notes
Day 10	Final Review and Submission (Phase 11)	All deliverables checked, packaged, and submitted

Note: This timeline is aggressive but achievable for a solo developer. For a team of 2–3, tasks on Days 4–7 can be parallelized (one member on supervised, one on unsupervised). Budget at least 2 additional days as buffer for unexpected issues.

End of Capstone Project Blueprint. This document serves as the complete reference for all phases, deliverables, technical decisions, and quality standards expected for the project. Refer to this document at every phase to ensure nothing is missed.